SIMATS ENGINEERING

ASSESSMENT TOOL 3 – ARTICLE REVIEW

Course Name: Artificial Intelligence   Course Code: CSA17

Course Outcome Covered: CO4 – Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making.

Article Selected

Kherif, O., Benmahamed, Y., Teguar, M., Boubakeur, A., & Ghoneim, S. S. M. (2021). *Accuracy Improvement of Power Transformer Faults Diagnostic Using KNN Classifier With Decision Tree Principle.* IEEE Access, 9, 81693–81701. DOI: 10.1109/ACCESS.2021.3086135.

Valid DOI: https://doi.org/10.1109/ACCESS.2021.3086135

Domain/Application: Electrical power systems, power-transformer condition monitoring and fault diagnosis.

ML Problem: Classification of faults in oil-immersed power transformers using dissolved gas analysis (DGA) measurements.

Article type: Peer-reviewed IEEE Access research article, published in 2021.

ARTICLE REVIEW

TASK 1: ARTICLE SUMMARY

(a) Citation details and problem

The selected research article was written by Omar Kherif, Youcef Benmahamed, Madjid Teguar, Ahmed Boubakeur, and Sherif S. M. Ghoneim. It was published in IEEE Access, Volume 9, pages 81693–81701, in 2021. The DOI is 10.1109/ACCESS.2021.3086135. The study belongs to the electrical engineering and artificial intelligence domain. Its application is power-transformer fault diagnosis. The specific machine-learning problem is to classify different fault conditions in oil-immersed power transformers from dissolved gas analysis (DGA) data. DGA is useful because faults inside transformer insulation can generate characteristic gases in the insulating oil. The authors focus on improving the classification accuracy by combining KNN with a decision-tree principle, rather than relying on a single conventional classifier.

(b) Objective, research gap and proposed solution (150–200 words)

The main objective of the article is to improve the accuracy of automatic power-transformer fault diagnosis using machine learning. Conventional DGA interpretation methods can have difficulty distinguishing fault types because different faults may produce overlapping gas patterns. The authors identify the need for a more accurate classification procedure that can select appropriate information for different stages of diagnosis. To address this gap, they combine the K-nearest-neighbour (KNN) classifier with a decision-tree principle. Instead of applying exactly the same input representation and classifier behaviour to every fault category, the proposed approach evaluates several DGA input vectors and selects suitable information for the corresponding decision node. The number of neighbours and distance measure used by KNN are also optimized according to classification performance. Two diagnostic models are developed and compared. The study uses 501 samples and additionally checks the proposed approach using an independent database. The reported global accuracy exceeds 93%, indicating that the proposed combination can improve transformer-fault classification compared with less structured approaches. Thus, the research contributes an interpretable, data-driven method for supporting transformer maintenance and condition monitoring.

TASK 2: METHODOLOGY ANALYSIS

(a) ML technique, dataset and preparation

KNN: K-nearest neighbours classifies a new sample by examining nearby training samples. The closest samples are selected according to a distance measure, and the class is inferred from their labels. In the article, the value of K and the distance type are optimized to improve diagnostic accuracy.

Decision-tree principle: The proposed diagnostic structure separates the problem into decision stages. Different input vectors can be selected at different nodes so that the classifier uses information most appropriate to the fault-separation task.

Dataset: The study uses 501 DGA samples for training/testing. The inputs are gas-concentration or gas-ratio based DGA vectors used for transformer-fault classification. Several alternative input vectors are assessed. The authors also use an independent database for complementary validation. The data are organized according to transformer fault categories and evaluated through classification accuracy.

(b) Mapping to CO4

The work maps directly to **Decision Trees** because a decision-tree principle controls the diagnostic stages. It also maps to **Statistical Learning / Classification** because KNN learns class boundaries from labelled examples and makes predictions for unseen samples. It demonstrates **Inductive Learning** because the system derives a general diagnostic rule from previously observed DGA examples.

Methodological strength: The method uses an independent database as complementary validation and optimizes KNN settings rather than selecting them arbitrarily.

Limitation: The dataset contains only 501 samples, which is relatively small for a complex industrial classification problem. Performance may therefore depend strongly on the representativeness of the available transformer cases and fault distributions.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key results

Model / Input	Accuracy (%)	Best distance	Best K
KNN – Vector 1	88.75	Cosine	6
KNN – Vector 2	91.88	Cityblock	10
KNN – Vector 3	78.13	Cityblock	8
KNN – Vector 4	73.13	Cityblock	7
KNN – Vector 5	32.50	Euclidean	39
KNN – Vector 6	63.13	Cosine	15
KNN – Vector 7	88.75	Euclidean	11
KNN – Vector 8	29.88	Cosine	44
KNN – Vector 9	90.63	Cityblock	6
Proposed overall diagnosis	>93	Optimized by node	Optimized

The article reports that the global accuracy of the proposed transformer diagnosis exceeds 93%. The individual KNN input-vector experiments show substantial variation, from 29.88% to 91.88%, demonstrating why selecting an appropriate representation at different decision stages is important.

(b) Critical evaluation

The reported accuracy above 93% is promising and is supported by the use of an independent validation database. However, accuracy alone is not sufficient for a safety-critical industrial diagnostic system. Transformer-fault classes may be imbalanced, so class-wise precision, recall, F1-score and a confusion matrix would provide a stronger picture of performance. The study's 501-sample dataset may not represent all transformer designs, operating ages, geographic conditions, oil properties and measurement noise. There is also a risk that high performance on a particular database may not transfer equally well to unseen transformer populations. A stronger evaluation would use repeated stratified cross-validation, report confidence intervals, test noisy and missing DGA measurements, compare against Random Forest, SVM and modern ensemble methods, and evaluate the system on data from multiple utilities.

(c) Real-world applicability

The technique can be deployed in transformer condition-monitoring systems, utility maintenance platforms and decision-support tools for electrical substations. It could help maintenance engineers prioritize inspections and identify likely fault categories from DGA measurements. Industry-scale deployment would face challenges including obtaining large, diverse and accurately labelled historical DGA datasets; differences between transformer manufacturers and operating environments; sensor and laboratory measurement noise; integration with existing monitoring systems; computational and maintenance costs; and the need for explainable decisions. Because transformer diagnosis can influence expensive maintenance decisions, the ML model should support—not blindly replace—qualified engineering judgement.

TASK 4: REFLECTION & LEARNING OUTCOME

(a) Three key insights

1. Feature/input selection affects classification quality. The large difference between the accuracies of the nine KNN input vectors shows that choosing informative features is essential. This connects to CO4 Statistical Learning and classification, where the quality of input representation directly affects model performance.

2. Decision trees can structure a difficult multi-class problem. Instead of forcing one classifier to distinguish every fault in the same way, decision stages can separate groups of faults progressively. This connects directly to the CO4 Decision Tree topic and demonstrates hierarchical decision-making.

3. Validation must test generalization. The independent-database validation is important because a model can perform well on its development data but fail on unseen data. This connects to Inductive Learning and model evaluation: the goal is to learn a general rule rather than memorize the training examples.

(b) Proposed extension

A useful extension would be to build a multi-utility transformer dataset containing DGA measurements from different transformer ratings, manufacturers, ages and operating environments. The proposed KNN-with-decision-tree approach could then be compared with Random Forest, SVM and gradient-boosting models using stratified cross-validation and an unseen utility as a final test set. In addition to accuracy, macro-F1, recall for each dangerous fault class, confusion matrices and calibration should be reported. The experiment is feasible because the existing method already works with DGA vectors, while the additional models are standard supervised-learning techniques. The expected impact is better evidence of generalization and a more reliable basis for industrial deployment.

CONCLUSION

The reviewed article demonstrates how KNN can be combined with a decision-tree principle to improve automatic power-transformer fault diagnosis. Its reported global accuracy above 93% and independent validation make the approach promising. At the same time, the relatively small dataset and reliance on accuracy indicate the need for broader datasets, class-wise metrics and stronger cross-domain validation. The study provides a clear practical example of CO4 concepts—inductive learning, decision-tree-based classification and statistical learning—being applied to a real engineering problem.

ARTICLE REVIEW – SUBMISSION DETAILS

Student Name: ______________________________

Register Number: ____________________________

Course: Artificial Intelligence (CSA17)

Assessment: Assessment Tool 3 – Article Review

Code of Conduct

I ____________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________________

Source / DOI

Kherif, O., Benmahamed, Y., Teguar, M., Boubakeur, A., & Ghoneim, S. S. M. (2021). Accuracy Improvement of Power Transformer Faults Diagnostic Using KNN Classifier With Decision Tree Principle. IEEE Access, 9, 81693–81701. DOI: 10.1109/ACCESS.2021.3086135.